

Predicting Water Potability: A Comparative Study of Decision Tree vs. Random Forest

1. Glossary of Key Terms

- **Stratified Split:** a partitioning method used to separate the data into training and testing sets. Aims to ensure the class ratio is kept consistent in both the train and test sets to prevent sampling bias.
- **Z-Score Normalisation:** standardises features by subtracting the mean and dividing by the standard deviation, resulting in the mean being 0 and the standard deviation being 1. This enables all variables to equally contribute to the model, even if they range in magnitude.
- **Overfitting:** when a function is too closely fit to a set of data points, it captures noise and instead of the signal, resulting in the training accuracy of the dataset being good, while the function performs poorly in predicting future dataset performance.
- **Cross-Validation (n-fold):** a statistical technique where a dataset is split into a number of (n) subsets and then the model is trained and tested on different combinations of $n - 1$ subsets, helping evaluate how well the model works on new data.
- **Decision Tree Pruning:** a method used for reducing overfitting by decreasing the size and complexity of a decision tree by eliminating the sections that provide little insight
- **Gini Impurity Index:** measures the amount of variety in a group (0 – 1). A score of 0 indicates a pure group where all samples belong to the same class; a higher score indicates the samples are a mix of different classes. The score is used to determine the best way to split the data in a decision tree.
- **Bagging/Bootstrapping:** Using different random samples of the data, multiple versions of a model are trained and then, through averaging all the results, a final model is created. This increases the accuracy of the model and reduces the chances of errors.
- **Out-of-Bag Error:** a method to test the accuracy of Random Forests. A mean prediction error is calculated on training samples using the trees that weren't included in the bootstrap training set.
- **F1-Score:** this evaluates how well a classification model performs on a dataset. In cases where the classes are imbalanced, where one class appears more than the other, this harmonic mean of precision and recall combines both metrics, creating a single value that balances their importance. (GeeksforGeeks, 2023)
- **Feature Importance:** the input variables are ranked by a score based on how much influence they had in making predictions. It indicates which factors the model relied on the most to make a decision.
- **Hyperparameters:** the different settings of the model that are tuned before the training to balance performance and prevent overfitting of the model.
- **Multicollinearity:** when two or more predictor (feature) variables are highly correlated, therefore, making it difficult to assess the effect of the individual features on the target variable.
- **Monotonic Ranking:** the variables tend to move together in the same way; for example, increasing or decreasing together relative to each other.
- **Probability Calibration:** The accuracy of the predictions made for a given class; for example, a prediction of 70% to be classified as "Potable" will correspond to the actual frequency of that class within the dataset.
- **Stochastic Error:** random variation or unpredictable noise can introduce errors into single estimator models as there are no established patterns to account for.
- **Type I Error:** a situation when the model predicts that a sample is Potable, when in reality, it is contaminated (false positive).
- **Type II Error:** a situation in which an individual predicts that a sample is contaminated, while in fact, it is Potable (false negative).
- **Greedy Search:** A strategy that uses algorithms as a way of choosing what is "best" at every small step(s) (for example, every time there is a "split" at some node) to eventually find the best overall solution

2. Intermediate Results and Analysis

2.1 Descriptive Statistics and Skewness

Features	Mean	Median	Std Dev	Skewness	Kurtosis
ph	7.086	7.0273	1.5733	0.04891	3.6186
Hardness	195.97	197.19	32.635	-0.085174	3.5229
Solids	21917	20934	8642.2	0.59545	3.3412
Chloramines	7.1343	7.1439	1.5848	0.012967	3.5472
Sulfate	333.22	332.23	41.205	-0.046523	3.7824
Conductivity	426.53	423.46	80.713	0.26667	2.7581
Organic_carbon	14.358	14.322	3.325	-0.020003	3.0294
Trihalomethanes	66.401	66.542	16.077	-0.051384	3.2211
Turbidity	3.9697	3.9682	0.78035	-0.033027	2.9478

Table 1: Descriptive Statistics

Solids and Conductivity are right-skewed, 0.59545 and 0.26667, respectively. Most of the features have high kurtosis values ($k>3$), which indicates heavy tails in comparison to the normal distribution where $k=3$

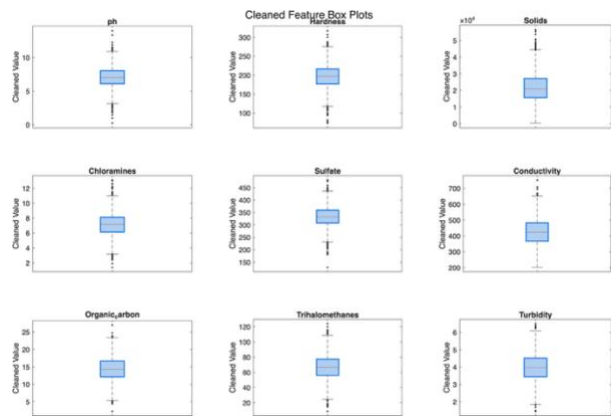


Figure 1: Feature Boxplots

Outlier Analysis (IQR Method)	
Variable	Number of Outliers
ph	46
Hardness	83
Solids	47
Chloramines	61
Sulfate	41
Conductivity	11
Organic_carbon	25
Trihalomethanes	33
Turbidity	19

Table 2: Number of Outliers in each Feature

The high number of outliers shown in the box plots for each Feature is statistically significant, as the statistical test for outliers was implemented by calculating the interquartile range (IQR) of each feature and counting the data points outside the IQR. As seen in the table above, which shows the significant number of outliers across all the Features in descending order, where ph has the highest number of outliers and Turbidity with the lowest.

2.2 Linear Independence and Class Separability

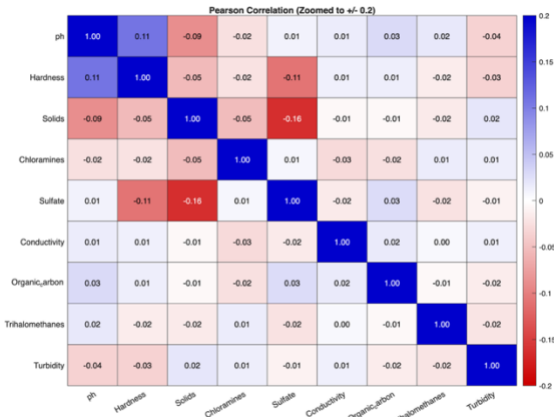


Figure 2: Feature Correlation Matrix (Zoomed)

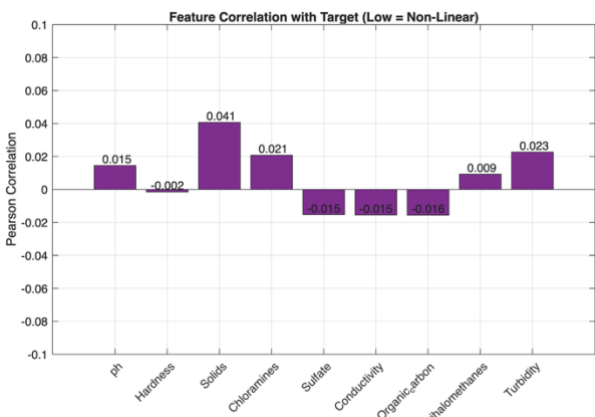


Figure 3: Feature Correlation with Target Bar Chart

The examination of intermediate relationships between features showed the high non-linearity of the data set and the weak dependencies between features and the target, with all feature inter-correlation coefficients remaining below 0.2 (Figure 2). This evidence of statistical independence from each other provided justification for the lack of concerns about multicollinearity; hence, the use of feature bagging through Random Forest is applicable here (due to the use of decorrelated predictors, Random Forest can effectively reduce variance). The target correlation overlaid histogram (Figure 6) depicts almost no linear dependence on each other, with the highest correlation equating to 0.041 (Solids).

Through inspection of the Overlaid Histograms (Figure 4), it was determined that the distribution of Potable was heavily nested inside the distribution of the Not potable class across all nine feature distributions, representing a significant negative result, showing that simple linear decision boundaries would not be adequate for classification purposes. Therefore, this evidence required moving away from simple linear decision boundary approaches and utilising more advanced, tree-based classifier architectures that perform well with respect to the complexities of multi-dimensional interactions.

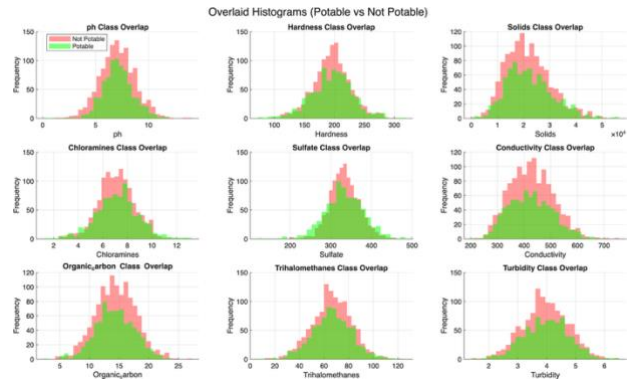


Figure 4: Overlaid Histograms for each Feature

2.3 Preliminary Performance & Negative Results

2.3.1 Data Attribution Impact

The raw dataset contained 1,434 missing values, which is a significant negative impact of data attrition. Deleting 1,265 rows (38.7% of the data) created a clean baseline; however, this deletion may also have caused the loss of some minor chemical patterns within the data. Therefore, this supports implementing MICE imputation in future work.

2.3.2 Baseline Decision Tre

The initial testing with a default Decision Tree resulted in a deep 22 level, 343 nodes structure that resulted in a 92.61% training accuracy but only a 59.20% accuracy on the test set. This large discrepancy in performance between training and testing accuracy, along with a baseline area under the curve of 0.56 and an F1-score of 0.48, indicated that the model was just memorising noise, thus performing only slightly better than random guessing. This leads to the mathematical justification for the pruning method and a hyperparameter tuning strategy used in refining generalisation for the final model.

Metrics	Results
Training Time	2.1502 sec
Tree Depth	22
Total Nodes	343
Training Accuracy	92.61%
Testing Accuracy	59.20%
F1-Score	0.4854
AUC	0.5646

Table 3: Default Decision Tree Metrics

3. Implementation Details

3.1 Strategic Preprocessing Choices

In order to maintain the integrity of the data, Z-score normalization was utilized in order to prevent high-magnitude features, such as solids, with a mean value of approximately 21,917, from numerically overwhelming lower-magnitude features, such as turbidity with a mean value of approximately 3.97. The model was validated using a Stratified 70/30 Partitioning approach. This technique guaranteed that the ratio of 59.7%/40.3% of classes was consistently present in the training and testing data, thus avoiding sampling bias when evaluating the model. Duplicates were also tested for and removed to eliminate the potential for data leakage during training.

3.2 Model Optimisation Strategies

3.2.1 Decision Tree

By applying 5-Fold Cross Validation, several distinct sets of MinLeafSizes were evaluated from 1 - 150 for an optimal growth of the tree structure. When analysing the results with the Decision Tree Optimisation Curve presented in Figure 5, it can be seen that the point where Cross Validation Error becomes lowest is at 20, which has been identified as being the preferred MinLeafSize for pruning trees further to reduce the significant overfitting observed when using only the average baseline MinLeafSize.

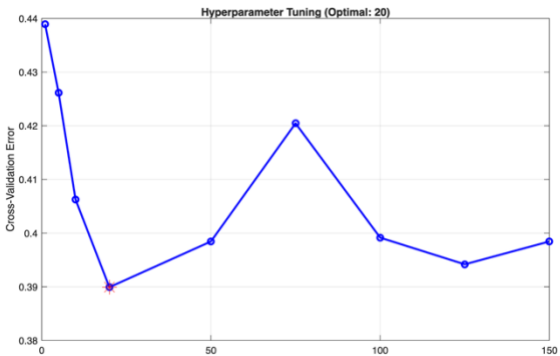


Figure 5: Hyperparameter tuning of Decision Tree

3.2.2 Random Forest

The Out-of-Bag (OOB) error estimation was used to identify the best size for the ensemble. The size of the ensemble was examined by using 50-200 trees in the search space. The RF Optimisation Curve (Figure 6) indicates that the predictability stabilised at about 200 trees.

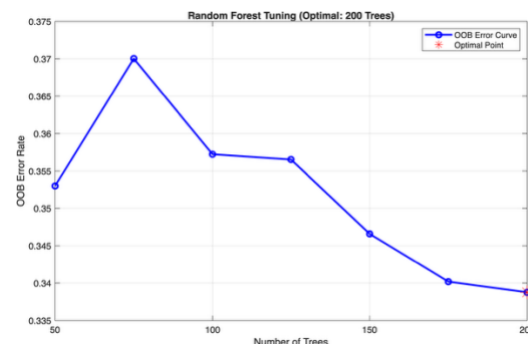


Figure 6: Hyperparameter tuning of Random Forest

3.3 Feature Importance

The comparative analysis of the feature importance ranking showed that each model emphasised chemical predictors differently. For example, the Decision Tree relied predominantly on pH when making greedy splits by using the Gini Importance score, as shown in Figure 7. The structural dependence of the Decision Tree indicates that one estimator could rely heavily on one linear separator, thus sacrificing its ability to capture layered and multi-dimensional relationships present in the dataset.

On the other hand, the Random Forest algorithm distributes the importance of features more evenly, with Sulfate and pH being the most important predictors of potability, as shown in Figure 8. This also illustrates one of the advantages of feature decorrelation through ensemble learning, since Random forests use random subsets of features for each tree to be able to see chemical patterns that the single Decision Tree could not. Random Forests also implemented Out-of-Bag Permuted Predictor Delta Error for the ensemble model, yielding a higher quality and unbiased assessment of which input features influence the model than traditional Gini Importance, which is used by the Decision Tree.

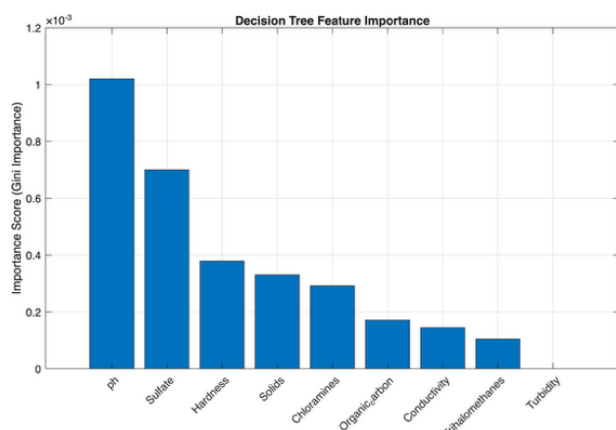


Figure 7: Decision Tree Feature Importance

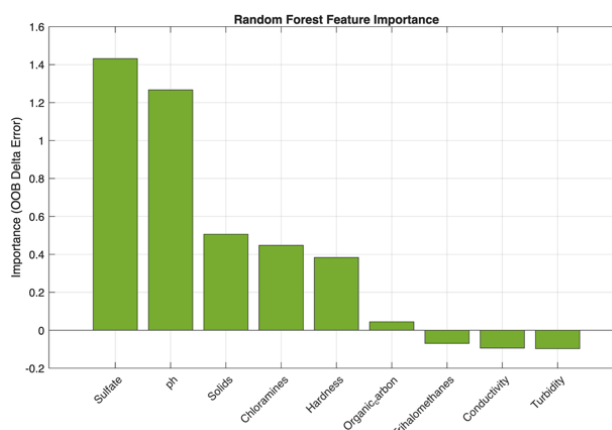


Figure 8: Random Forest Feature Importance